

Formal Verification of Imperative First-Class Functions in Move

FMCAD 2026

Wolfgang Grieskamp Teng Zhang Vineeth Kashyap Jake Silverman

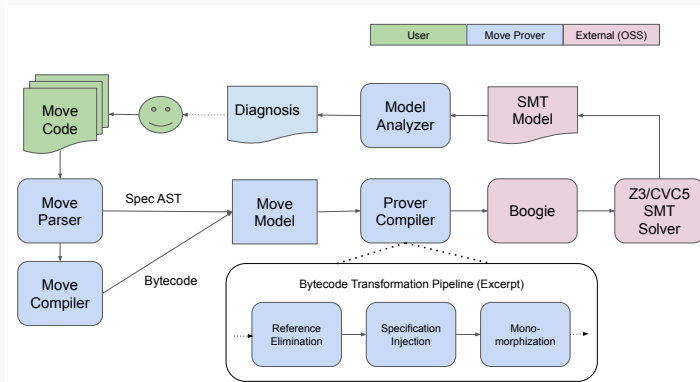
Aptos Labs, Palo Alto, USA

The Move Prover

From Meta to Aptos Labs

Specs first-class
citizens of code

Verifies bytecode
after the compiler



Detailed
diagnostics,
supporting AI
feedback loop

Fast and reliable
encoding,
runs in CI
(TACAS'22)

Move went higher-order

- Function types $|T_1, \dots, T_n|T$, closures with captured values.
- A closure can be part of a struct and be persisted in global storage.
- Persisted function values are Move's principled account of **dynamic dispatch**.

Use cases: callbacks, extensible asset managers, dispatchable permissions, fold/map/filter.

```
// Liquidity pool of an automated market  
// maker (AMM), as in Uniswap v2 (Ethereum)  
struct Pool {  
    reserve_x: u64, reserve_y: u64,  
    // (reserve_in, reserve_out, amt) = out  
    pricing: |u64, u64, u64|u64  
}  
  
// Extract of the AMM's core functionality  
fun swap(pool: &mut Pool, amt: u64): u64 {  
    let out = (pool.pricing)(  
        pool.reserve_x, pool.reserve_y, amt);  
    pool.reserve_x += amt;  
    pool.reserve_y -= out;  
    out  
}
```

The challenge

MVP is modular: a call site looks up the callee's spec. With dynamic dispatch there is no callee to look up.

This paper

1. **Behavioral predicates:** $\text{aborts_of}\langle f \rangle(\bar{x})$, $\text{ensures_of}\langle f \rangle(\bar{x}, \bar{y})$, $\text{result_of}\langle f \rangle(\bar{x})$ allow to reason about some function value f
2. **State labels:** name intermediate states, $S_1..S_2 \mid \sim \phi$.
3. **Encoding:** discriminate over the function values that can reach a call site.
4. **Specification inference** for higher-order code.

Running example: what may a pricing function do?

```
spec Pool {  
  reads_of<self.pricing> *;  
  
  invariant  $\forall S \text{ in } *, ri: u64, ro: u64, a: u64:$   
     $S \models \neg \text{aborts\_of}\langle\text{self.pricing}\rangle(ri, ro, a);$   
  
  invariant  $\forall S \text{ in } *, ri: u64, ro: u64, a: u64:$   
     $S \models \neg \text{result\_of}\langle\text{self.pricing}\rangle(ri, ro, a) \leq ro;$   
  
  invariant  $\forall S \text{ in } *, ri: u64, ro: u64,$   
     $a1: u64, a2: u64:$   
     $S \models a1 \leq a2 \implies$   
       $\text{result\_of}\langle\text{self.pricing}\rangle(ri, ro, a1)$   
       $\leq \text{result\_of}\langle\text{self.pricing}\rangle(ri, ro, a2);$   
  
  invariant  $\forall S \text{ in } *, ri: u64, ro: u64, a: u64:$   
     $S \models (ri + a)$   
       $* (ro - \text{result\_of}\langle\text{self.pricing}\rangle(ri, ro, a))$   
       $\geq ri * ro;$   
}
```

A data invariant on Pool constrains *any* closure stored in pricing:

1. never aborts,
2. output bounded by output reserve,
3. monotone in the input,
4. product of reserves never decreases.

`aborts_of`, `result_of`: behavior of the stored function value; `S`: a state label over all reachable states.

Checked when a Pool is built, assumed wherever a Pool is used.

Behavioral predicates

For a function value f with arguments $\bar{x}$ and results $\bar{y}$:

<code>requires_of</code> $\langle f \rangle(\bar{x})$	precondition holds
<code>aborts_of</code> $\langle f \rangle(\bar{x})$	the call aborts
<code>ensures_of</code> $\langle f \rangle(\bar{x}, \bar{y})$	postcondition holds (two-state)
<code>result_of</code> $\langle f \rangle(\bar{x})$	the result of the call (two-state)
<code>reads_of</code> $\langle f \rangle R$	memory the call may read
<code>modifies_of</code> $\langle f \rangle(\bar{x}) R[a]$	memory the call may write

- On a known function they unfold to its spec.
- On a parameter or a stored closure they stay abstract.

State labels

- $S \models p$: evaluate p in state S .
- $S_1 \models q$: S is the pre-state of a call; $S_1..S_2 \models q$ names both ends.
- $\forall S \text{ in } *$ for invariants, $\exists S \text{ in } *$ to witness an intermediate state:

```
fun followed_by(f: |&mut A|,
               g: |&mut A|,
               x: &mut A) {
  f(x); g(x)
}
spec followed_by {
  reads_of<f> *; reads_of<g> *;
  ensures  $\exists S \text{ in } *$ :
    ( $S \models \text{ensures\_of}<f>(x)$ ) &&
    ( $S.. \models \text{ensures\_of}<g>(x)$ )
}
```

The post-state of f becomes the pre-state of g , without threading values through the spec.

Application side: swap

```
spec swap {  
  aborts_if pool.reserve_x + amt > MAX_U64;  
  ensures result = old(result_of<pool.pricing>(pool.reserve_x, pool.reserve_y, amt));  
  ensures pool.reserve_x * pool.reserve_y ≥ old(pool.reserve_x * pool.reserve_y);  
}
```

- Pricing cannot abort by (1), reserve_y - out cannot underflow by (2): only the overflow abort remains.
- Product preservation follows from (4).

With dynamic dispatch, the application side is no harder than first-order code.

Compliance side: building a pool

```
// fee-free constant product: verifies
fun product(ri: u64, ro: u64, a: u64): u64 { .. }

struct Fee {bps: u64}
fun with_fee(owner: address,
             ri: u64, ro: u64, a: u64): u64 {
    let b = (a as u128)
        * (10000 - Fee[owner].bps as u128) / 10000;
    product(ri, ro, b as u64)
}
spec with_fee {
    aborts_if !∃<Fee>(owner);
    aborts_if Fee[owner].bps > 10000; ..
}
Pool{..,
    pricing: |ri, ro, a| with_fee(owner, ri, ro, a)}
// error: data invariant does not hold
```

Storing f where p is expected requires, for all inputs,

$$\begin{aligned} \text{requires_of}\langle p \rangle(\bar{x}) &\Rightarrow \text{requires_of}\langle f \rangle(\bar{x}) \\ \text{aborts_of}\langle f \rangle(\bar{x}) &\Leftrightarrow \text{aborts_of}\langle p \rangle(\bar{x}) \\ \text{ensures_of}\langle f \rangle(\bar{x}, \bar{y}) &\Rightarrow \text{ensures_of}\langle p \rangle(\bar{x}, \bar{y}) \end{aligned}$$

- with_fee violates the no-abort invariant; a guarded variant with a default fee verifies.
- The nonlinear fee variants need a one- or two-line proof hint.

Encoding: which function values can reach a call site?

After monomorphization this is known statically. Per function type τ , one datatype with a constructor per source:

```
datatype  $\tau$  {  
     $C_f(\overline{c : T}), \dots, P_{f,x}(), \dots, F_{\sigma.x}(n : \mathbb{N}), \dots$   
}
```

- $C_f(\overline{c})$: closure of function f with captured values $\overline{c}$.
- $P_{f,x}$: function-typed parameter x of a verification target f .
- $F_{\sigma.x}(n)$: struct field x of σ ; the discriminator n lets two locations hold distinct closures.

A VC only sees the function values derivable from the program.

Encoding: invoking a function value

```
proc  $invoke_\tau(x : \tau, \overline{p : T})$  returns  $(\overline{r : T})$   
  if  $x$  is  $C_f(\overline{c})$   
     $\overline{r} := \text{call } f(\overline{c}, \overline{p})$   
  else  
    assert  $requires_\tau(x, \mathcal{M}_\tau, \overline{p})$   
    if  $aborts_\tau(x, \mathcal{M}_\tau, \overline{p})$   
       $abort\_flag := \text{true}$   
    else  
      let  $\mathcal{M}'_\tau := \mathcal{M}_\tau$   
      havoc  $\Delta_\tau(\overline{p}), \overline{r}$   
      assume  $ensures_\tau(x, \mathcal{M}'_\tau, \mathcal{M}_\tau, \overline{p}, \overline{r})$ 
```

- Known closure: call the function.
- Otherwise: the usual opaque-call schema, driven by the behavioral predicates.
- $\mathcal{M}_\tau, \Delta_\tau$: read and write footprints of τ .

Encoding: intermediate state labels

```
struct R(u64);  
spec foo(f: |address|, g: |address|,  
        a: address) {  
  aborts_if aborts_of<f>(a);  
  ensures ..S |~ ensures_of<f>(a);  
  ensures S |~ R[a].0 > 0;  
  aborts_if S |~ aborts_of<g>(a);  
  ensures S.. |~ ensures_of<g>(a);  
}
```

The state at S becomes an existentially quantified memory snapshot $\mathcal{M}^S$.

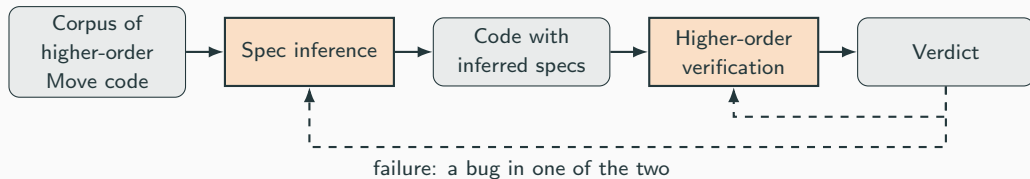
```
fun abortsCfoo[ $\mathcal{M}', \mathcal{M}$ ](a)  
  aborts $\tau$ (f,  $\mathcal{M}', a$ )  
   $\vee (\exists \mathcal{M}^S \bullet \text{ensures}_{\tau}(f, \mathcal{M}', \mathcal{M}^S, a)$   
     $\wedge R[\mathcal{M}^S][a].0 > 0$   
     $\wedge \text{aborts}_{\tau'}(g, \mathcal{M}^S, a))$ 
```

An abort at S needs S established first: the ensures reaching S are conjoined with the abort condition; pre-state aborts are disjoined in front.

Sound since a verified function satisfies $\bigvee \bar{A} \vee \bigwedge \bar{P}$.

Specification inference as a test oracle

MVP also has the reverse path: WP-based *specification inference* derives specs from code, using behavioral predicates for calls. Inferred specs are complete and are expected to verify.



Two independent algorithms are unlikely to agree on a bug. The current test suite covers 550 verification conditions.

- **Dafny**: behavioral predicates and labels, but function values cannot modify state, and labels are not quantified.
- **Verus, Prusti**: closure specs by state passing or separation logic; Move's alias-free references and resource-separated memory avoid both.
- **F***: effectful higher-order code with WP in the type system; heap separation is reasoned about in the logic, whereas Move separates memory at compile time.
- **Solidity verifiers** (Certora and others): dispatch by enumerating callees; our datatype makes the case split explicit and falls back to behavioral predicates.

Conclusion

- Behavioral predicates and state labels make function specs first-class.
- The encoding discriminates on the variant reaching a call site: concrete effect when known, behavioral predicates otherwise.
- MVP's routine, counterexample-driven verification now covers dynamic dispatch; inference validates the encoding.

Future work

- Quantifying over *sequences* of state labels for fold-style contracts.
- Better automation for the nonlinear compliance proofs.
- Encoding and verification of the schema in Lean.

Thank you

Prover, examples, and tests: github.com/aptos-labs/aptos-core

Paper examples: https://github.com/aptos-labs/aptos-core/tree/28f82080e3/third_party/move/move-prover/doc/higher-order-paper-26/examples

Backup: higher-order loops, find

```
fun find<T>(v: &vector<T>, pred: |&T|bool): u64 {
  let i = 0; let n = v.length();
  while (i < n) {
    if (pred(v[i])) return i;
    i += 1;
  } spec {
    invariant i ≤ n;
    invariant no_match_before(v, pred, i);
  };
  n
}
spec find {
  ensures result ≤ len(v);
  ensures no_match_before(v, pred, result);
  ensures result < len(v)
    ⇒ result_of<pred>(v[result])
}
```

```
spec fun no_match_before<T>(v: vector<T>,
  pred: |&T|bool, end: u64): bool
{ ∀ j in 0..end: !result_of<pred>(v[j]) }

fun find_zero_lambda(v: &vector<u64>): u64 {
  find(v, |x| x = 0
    spec { ensures result = (x = 0) })
}
spec find_zero_lambda {
  ensures result < len(v) ⇒ v[result] = 0
    && (∀ j in 0..result: v[j] ≠ 0);
  ensures result = len(v)
    ⇒ (∀ j in 0..len(v): v[j] ≠ 0);
}
```

The loop invariant is written once; callers verify from find's spec and the lambda's contract.

Backup: memory access sets

```
spec foo(f: |address|, g: |T|S) {  
  reads_of<f> R;           // f may read resource R  
  reads_of<g> *;           // g may read anything in scope, plus unknown memory  
  modifies_of<f>(a: address) R[a]; // f may write R[a]; default is no writes  
}
```

- $\mathcal{M}_\tau = \mathcal{M}_{C_f} \cup \mathcal{M}_{P_{f,x}} \cup \mathcal{M}_{F_{\sigma,x}}$: derived from code for closures, declared via `reads_of` for parameters and fields.
- `reads_of<g> *` adds an abstract map for unknown memory, so $g(x)$ is not assumed equal across states.
- Δ_τ likewise, from `modifies` and `modifies_of`; no wildcard, since it would leave no frame.

Backup: encoding determinism

$ensures_\tau$ switches over the variant and adds the frame condition (backup). It is a relation; Move functions are deterministic.

Introduce an uninterpreted $result_\tau$ and tie it to $ensures_\tau$:

```
fun  $result_\tau[\mathcal{M}', \mathcal{M}](x : \tau, \overline{p} : \overline{T}) \rightarrow \overline{U}$   
axiom  $\forall x, \overline{p} \bullet \{result_\tau[\mathcal{M}', \mathcal{M}](x, \overline{p})\}$   
    let  $\overline{r} := result_\tau[\mathcal{M}', \mathcal{M}](x, \overline{p}) \bullet ensures_\tau[\mathcal{M}', \mathcal{M}](x, \overline{p}, \overline{r})$ 
```

- One direction only: a biconditional would force all solutions of a weak ensures to coincide.
- For parameter and field variants, *ensures* is *defined* as the graph of an uninterpreted *result*.

Backup: ensures with frame condition

The caller havoced all of Δ_τ ; the variant only constrains Δ_x :

```
fun ensures $_\tau$ [ $\mathcal{M}'$ ,  $\mathcal{M}$ ]( $x : \tau, \overline{p : T}, \overline{r : U}$ ) : bool
   $\bigwedge_{(\rho, a) \in \Delta_\tau}$  (
    if  $\rho \notin \mathcal{T}(\Delta_x)$ 
       $\mathcal{M}(\rho) = \mathcal{M}'(\rho)$ 
    else
      ( $\bigwedge_{(\rho, a') \in \Delta_x} a' \neq a$ )
        $\Rightarrow \mathcal{M}(\rho)[a] = \mathcal{M}'(\rho)[a]$ 
  )  $\wedge$  ensures $_x$ [ $\mathcal{M}'_x$ ,  $\mathcal{M}_x$ ]( $p, r$ )
```

Modification addresses are symbolic and may alias, hence the pairwise disequalities.

Backup: making it fast

- Predicates as IVL functions *with bodies* time out: Boogie unfolds eagerly.
- Instead: *uninterpreted* predicates, connected by axioms with per-variant triggers.

axiom $\forall \bar{c}, \bar{p}, \bar{r} \bullet \{ \text{ensures}_\tau(C_f(\bar{c}), \bar{p}, \bar{r}) \}$
 $\text{ensures}_\tau(C_f(\bar{c}), \bar{p}, \bar{r}) \Leftrightarrow \text{ensures}_{C_f(\bar{c})}(\bar{p}, \bar{r})$
axiom $\forall \bar{p}, \bar{r} \bullet \{ \text{ensures}_\tau(P_{f,x}(), \bar{p}, \bar{r}) \}$
 $\text{ensures}_\tau(P_{f,x}(), \bar{p}, \bar{r}) \Leftrightarrow \text{ensures}_{P_{f,x}}(\bar{p}, \bar{r})$
axiom $\forall f, \bar{p}, \bar{r} \bullet \{ \text{ensures}_\tau(f, \bar{p}, \bar{r}) \}$
 $f \text{ is } F_{\sigma.x} \Rightarrow (\text{ensures}_\tau(f, \bar{p}, \bar{r}) \Leftrightarrow \text{ensures}_{F_{\sigma.x}}(f.n, \bar{p}, \bar{r}))$

Axioms fire only where the variant is syntactically present in the proof context.

Backup: inference test categories

Category	Representative cases
Direct calls	Chained calls to a spec'd helper; inferred specs use <code>ensures_of</code> and <code>aborts_of</code> instead of inlining the body.
Mutual recursion	<code>is_even</code> and <code>is_odd</code> ; each spec is phrased via the other's behavioral predicate.
Higher-order parameters	Closure applied by the callee; spec abstracts via predicates on the parameter.
Stored-closure dispatch (enum)	Calculator with a closure-carrying enum variant.
Stored-closure dispatch (field)	Vault with a <code>Strategy</code> closure field on Aptos framework types; <code>modifies_of</code> inferred for the field.
Pre/post state labels	<code>move_to</code> , <code>move_from</code> , mutable resource indexing; <code>@pre</code> versus post-state memory.
Intermediate state labels	Sequenced state-modifying calls; existentially quantified labels pinned to defining predicates.

Backup: how the implementation was built

Developed in close collaboration with AI coding assistants (primarily Claude Code; Codex as a second opinion).

Went well

- Frontend extensions largely AI-produced, human review only.
- Root-cause diagnosis once inference surfaced a failing example.
- Timeout diagnosis via Boogie, SMT, and Z3 instantiation logs.

Went less well

- The encoding itself needed substantial human guidance; soundness calls were human.
- Dozens of iterations in the hard parts; effort approached manual implementation.
- Token cost.

Key enabler: the inference-then-verify pipeline gave an unambiguous correctness signal independent of human review.